

UC3: Retract Bid

Primary Actor: Bidder

Stakeholders and Interests:

- *Bidder*: wants a fair way to undo a genuine mistake (e.g., typo) without it feeling like a loophole
- *Seller*: wants confidence that bids are generally binding and can't be casually cancelled to manipulate the auction
- *System*: must enforce a strict time window and prevent abuse of the retraction feature

Preconditions:

- Bidder is logged in
- Bidder has an active bid on the item
- Auction is still open (not yet closed)

Postconditions (success):

- The retracted bid is marked as withdrawn and no longer counts as the current highest bid
- The next-highest valid bid (if any) becomes the new current highest bid
- The retraction is logged in the audit trail with timestamp and reason
- If no other bids exist, the current price reverts to the starting price (or the next-highest remaining bid)

Main Success Scenario:

1. Bidder navigates to their dashboard and selects the active bid they want to retract.
2. System checks that the bid was placed within the last 15 minutes.
3. System checks that the auction is still open.
4. System prompts the bidder to confirm the retraction and briefly state it was a mistake.
5. Bidder confirms.
6. System marks the bid as withdrawn.
7. System recalculates the current highest bid from remaining valid bids.
8. System logs the retraction (bidder, original amount, timestamp, retraction timestamp) in the audit trail.
9. System updates the live price display for all viewers in real time.
10. System confirms to the bidder that the bid was successfully retracted.

Extensions (Alternate Flows):

- **2a. Bid was placed more than 15 minutes ago:**
 1. System rejects the retraction request.

2. System displays: "This bid can no longer be retracted — the 15-minute window has passed."
 3. Use case ends.
- **3a. Auction has already closed:**
 1. System rejects the retraction request.
 2. System displays a message that closed auctions cannot be modified.
 3. Use case ends.
 - **6a. Bidder attempts to retract a bid that is no longer their highest (i.e., they've already been outbid):**
 1. System still allows the retraction (it's a withdrawal of their own bid record, not a price change), but notes it has no effect on the current price since it wasn't the leading bid.
 2. System logs the retraction for audit purposes.
 3. Use case ends.
 - **6b. Bidder attempts a second retraction on the same item:**
 1. System checks retraction history for this bidder/item pair.
 2. System rejects if the bidder has already used their retraction for this bid (retraction is one-time per bid, not per item, to prevent repeated gaming).
 3. System displays a message explaining the limit.

Special Requirements:

- Retraction window is a hard 15-minute limit measured from the original bid timestamp, not from the retraction request time — important to model precisely for your timing tests.
- Retraction does **not** delete the bid record; it's a status change (**active** → **withdrawn**), preserving full audit history.
- This use case is a secondary but useful target for boundary-value testing (e.g., testing retraction attempts at 14:59 vs 15:00 vs 15:01 after the bid).